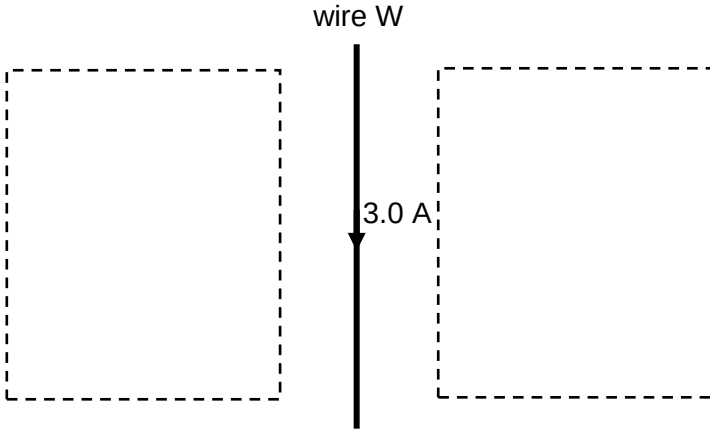
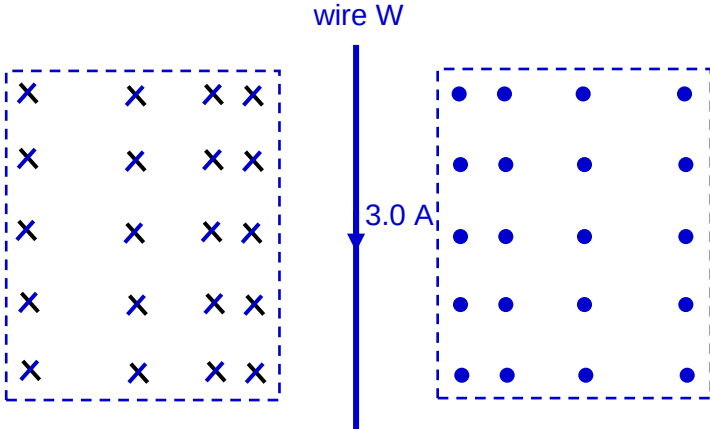
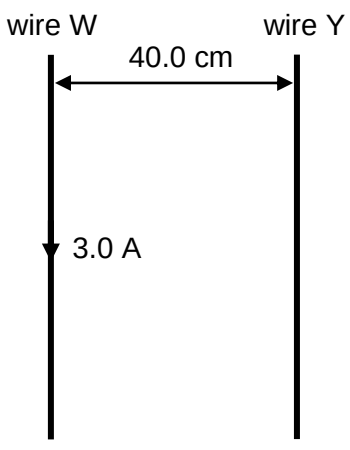


2	A long, straight wire W carrying a direct current of 3.0 A flows in the direction as shown in Fig. 2.1.  Fig. 2.1
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(a)	Draw on Fig. 2.1, the pattern of the magnetic field produced by wire W in the regions indicated by the dotted boxes. Use the symbol x to represent magnetic field directed into the page and use the symbol • to represent magnetic field directed out of the page.	[3]
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L1	Solution:  Marks scheme: 1 mark for: - Correct direction of magnetic flux density on left side (into plane of paper) and right side (out of plane of paper) of wire W. 1 mark for: - Increasing spacing with increasing distance from wire W. 1 mark for (Pattern shown sufficiently) - At least 4 columns each of x and • shown to show the trend of increasing spacing between consecutive circular field lines.	B1 B1 B1
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		At least 3 rows each of x and • shown to show the trend of equal spacing parallel to the wire.	
	(b)	A similar wire Y is placed parallel to wire W, separated by a distance of 40.0 cm as shown in Fig. 2.2. Initially, there is no current in wire Y.  Fig. 2.2	
	(i)	Show that the magnetic flux density at wire Y due to the current in wire W is $1.5 \times 10^{-6} \text{ T}$.	
			[1]
	L1	Magnetic flux density $B = \frac{\mu_0 I}{2\pi d}$ where I is the current in wire W and d the distance of Y from W. $B = \frac{(4\pi \times 10^{-7})(3.0)}{2\pi(40.0 \times 10^{-2})}$ $= 1.5 \times 10^{-6} \text{ T (Shown)}$	M1 A0
	(ii)	A current of 1.0 A is now switched on in wire Y and flows in the opposite direction as the direction of current flow in wire W. Use your answer in (b)(i) to calculate the force per unit length acting on wire Y.	
		force per unit length = N m^{-1}	[2]
	L2	Magnetic force $F_{\text{on Y}} = B_{\text{at Y due to W}} I_{\text{in Y}} L_{\text{of Y}} \sin \theta$ where $\theta = 90^\circ$ since B and I are perpendicular. $\frac{F}{L} = BI$ $\frac{F}{L} = (1.5 \times 10^{-6})(1.0)$ $= 1.5 \times 10^{-6} \text{ N m}^{-1}$	M1 A1
	(iii)	Explain why the force that the two wires exert on each other is repulsive.	
			
			
			

				
				
				
				[3]
		L2	By Right Hand Grip Rule (or, using Fig. 2.1), the magnetic field produced by the current in wire W acts <u>perpendicular</u> to wire Y and <u>out of</u> the plane of paper. By Fleming's Left Hand Rule, the direction of the magnetic force <u>on Y by W</u> acts <u>to the right</u> / away from wire W. By Newton 3rd law of motion, the direction of the magnetic force <u>on W by Y</u> is <u>opposite to that on wire Y</u> / towards the left / away from wire Y. [OR, Apply Fleming's Left Hand Rule a second time to determine the force on W.] Therefore there is a repulsive force acting between the two wires.	M1 M1 M1 A0
		(iv)	Determine a possible position , other than at infinity, where the resultant magnetic flux density due to the magnetic fields of both wires is zero .	
			position:	[3]
		L2	First, consider directions of the two fields in different regions: - Net B cannot be zero between W and Y, since their fields point in the same direction. <u>On the left of both wires, and, on the right of both wires, their fields point in opposite directions.</u> Secondly, consider magnitude of B: - Since $B = \frac{\mu_0 I}{2\pi d} \Rightarrow B \propto \frac{I}{d}$. Hence Net B can only be zero <u>on the side further away from the larger current</u>. Thus, Net B is zero on the <u>right side of wire Y</u>. Let y be the distance to the <u>right of wire Y</u>.	M1 M1

		$ B_{\text{due to Y}} = B_{\text{due to W}} $ $\frac{\mu_0 (1.0)}{2\pi y} = \frac{\mu_0 (3.0)}{2\pi (40.0 \times 10^{-2} + y)}$ $\frac{(1.0)}{y} = \frac{(3.0)}{(40.0 \times 10^{-2} + y)}$ $40.0 \times 10^{-2} + y = 3.0y$ $2.0y = 40.0 \times 10^{-2}$ $y = 20.0 \times 10^{-2} \text{ m} = 20.0 \text{ cm}$ position: <u>20.0 cm to the right of wire Y.</u> <i>(For marking: 'right' of wire Y can be stated either in the final answer space, or, working. Award A1 mark as long as this understanding that it must be right of Y be seen.)</i>	A1
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[Total: 12]

2	A uniform spherical star has a mass of $6.0 \times 10^{30} \text{ kg}$. The mass of the star may be assumed to
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